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人形机器人

人形机器人前沿：莫拉维克悖论

我们在2026世界人形机器人运动会中看到了显着进步，但行业关注的核心问题仍是：机器人能否高效且有生产力地完成真实任务。近期新的机器人模型显示出规模定律 (Scaling Law) 的早期信号，商业化落地进展仍是投资者最关注的方向。

“让计算机在智力测试中达到成年人的水平相对容易；而要赋予其类似一岁婴儿的感知和运动能力，则十分困难，甚至是不可能的。”—— 汉斯·莫拉维克 (Hans Moravec)，1988年。行业现时已逐步解决了机器人的运控问题。那么，感知能力以及对物理世界的推理能力进展如何？

八月亮点：

- 2026世界人形机器人运动会：今年共吸引666支队伍和2,056台机器人参赛（2025年约为280支队伍）。机器人在运动能力方面的突破令人印象深刻，多项人类纪录被大幅刷新。例如，天工机器人完成100米跑仅需8.64秒，不仅显著快于其2025年的21.5秒，也超过了博尔特9.58秒的人类世界纪录。在应用场景方面，与2025年相比，2026年赛事更强调真实环境中的多步骤长序列任务执行，而非单一动作。21项场景赛覆盖工厂、家庭、酒店、零售、医院及应急响应等领域。以餐饮场景为例，机器人需要理解语音指令，取出并加热食物，领取饮料，随后将餐品送达指定位置。场景赛的评分规则明确鼓励自主执行能力，完全自主权重是遥控的两倍。此外，本届赛事首次设立灵巧手专项赛，涵盖八项精细操作任务，包括称量粉末、搭建积木、镊子夹豆、线缆连接等。
- 大脑和数据仍是全球行业焦点。近期，美国机器人实验室相继推出DYNA-2、Skild AI S1和Generalist GEN-1.5等新模型，重点提升机器人学习新任务的效率，降低教授机器人新任务（后训练）的边际成本。据介绍，上述模型已能够通过少量示例学习完成新任务，并具备执行长序列复杂任务的能力。与此同时，过去以硬件见长的企业也正将更多资源投入机器人智能系统研发。宇树科技计划将IPO募集资金的48%用于机器人模型研发；众擎机器人发布具身智能引擎Awaken，是一套仿人脑五层体系架构。数据依然是行业发展的主要瓶颈。我们认为当前行业正逐步转向第一视角视频或UMI数据。中国企业受益于较低的人力成本和更广泛的数据来源，Figure也推出Figure Index，通过向手机应用用户付费的方式收集第一视角的人类任务视频，用于训练其Helix模型。公司计划未来12个月投入10亿美元用于数据采集。

This translated report is made available for convenience only and is excerpted from the original research report published in English. In the event of any discrepancy between the translation and the original research report, the content in the original research report will prevail. The original research report can be found here: [Humanoids: Humanoid Horizons: Moravec's Paradox \(1 Sep 2026\)](#).

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本报告内容还包括：

- 近期人形机器人更新与进展
- 中国政策支持概览
- Humanoid 100表现
- 人形机器人进展跟踪
- 全球人形机器人TAM/采用率估算

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进一步评论



Tech Diffusion

A Morgan Stanley Research
Key Theme of 2026

世界机器人大会2026要点——应用扩大，生产力更重要：与去年相比，本届大会的变化不再主要体现在运控能力的再次跃升，而是更关注机器人能否完成有用任务、适应环境变化，并逐步接近人类生产力。中国人形机器人的商业化应用正在拓宽，更多客户从试点转向小批量部署，但多数场景的部署规模仍只有数十台。公司交流的核心集中在机器人智能。VLA、世界模型与强化学习（RL）正越来越多地结合使用；相比单一架构，数据配方和部署反馈闭环看起来更具持续性。数据仍是瓶颈，但数据良率、多样性和部署反馈的重要性正在上升。各家公司计划重点提升：(1) 模型参数规模；(2) 算力（已有多家企业达到约千卡规模）；(3) 数据采集（目标在2026年底或2027年达到100万小时可用数据）。

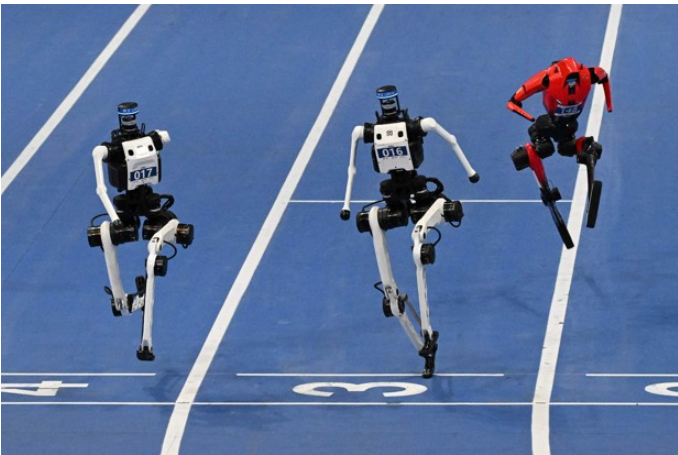
- 详见我们的详细报告：[WRC 2026 Takeaways](#)

Exhibit 1: 世界人形机器人运动会奖牌榜

Rank	Integrators	Gold	Silver	Bronze	Total
1	-	18	16	12	46
2	Tiangong	15	12	18	45
3	Unitree	3	4	3	10
4	Booster Robotics	3	3	3	9
5	Galbot	2	1	1	4
6	Lumos	1	1	0	2
7	UniX	1	1	1	3
8	Spirit AI	1	1	0	2
9	Noetix	1	0	1	2
10	Galaxea	1	0	0	1
10	Fairino	1	0	0	1
10	Qijiang	1	0	0	1

资料来源：世界人形机器人运动会、摩根士丹利研究部。注：仅按金牌数量列示前10名。

Exhibit 2: 天工100米跑出8.64秒，打破博尔特9.58秒的人类世界纪录



资料来源：CGTN

Exhibit 3: 除田径竞赛外，本届赛事还设置了灵巧手专项赛



资料来源：盖世汽车（Gasgoo）

Exhibit 4: 与单一预设动作相比，2026年赛事更重视真实场景多步骤任务



资料来源：环球时报（Global Times）

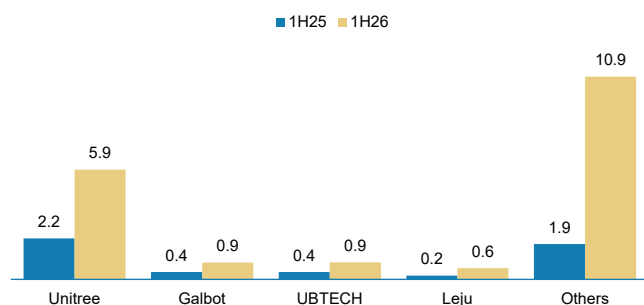
2026年上半年人形机器人销量快速增长：全球出货量达到1.9万台，同比增长272%，其

中中国厂商占97%。尽管中国约65%的出货仍用于娱乐、教育/研发和数据采集，但应用正在向工业和商业场景拓展。按形态看，半尺寸双足机器人仍是销量最大的类别（约50%），轮式机器人在商业和工业场景中快速扩张（约29%）。

- 我们预计，2026年上半年的更多试点项目将在下半年转化为更大范围部署，并预测中国人形机器人出货量将在2026年达到5万台，2030年达到44.6万台。

Exhibit 5: 根据SAG数据，2026年上半年全球人形机器人出货量达到1.9万台，2025年上半年为0.5万台

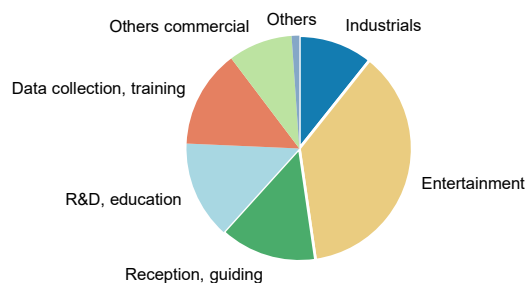
Global Humanoid Shipment ('k unit)



资料来源：Smart Analytics Global、摩根士丹利研究部

Exhibit 6: 从行业层面看，我们在2026年上半年看到更多商业应用场景……

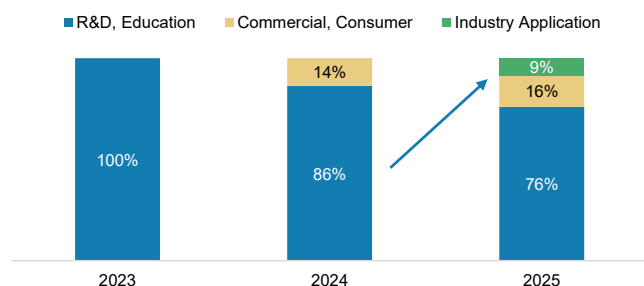
1H26 China Humanoid Shipment by Applications



资料来源：MIR Databank、摩根士丹利研究部

Exhibit 7: ……相比宇树在2025年的收入高度偏向研发与教育

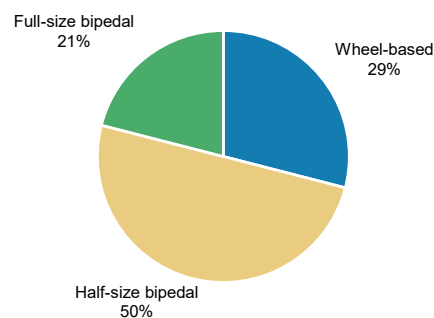
Unitree: Humanoid Revenue Breakdown by Use Cases



资料来源：宇树、摩根士丹利研究部

Exhibit 8: 2026年上半年——中国人形机器人按类型划分的出货量

1H26 China Humanoid Shipment by Type

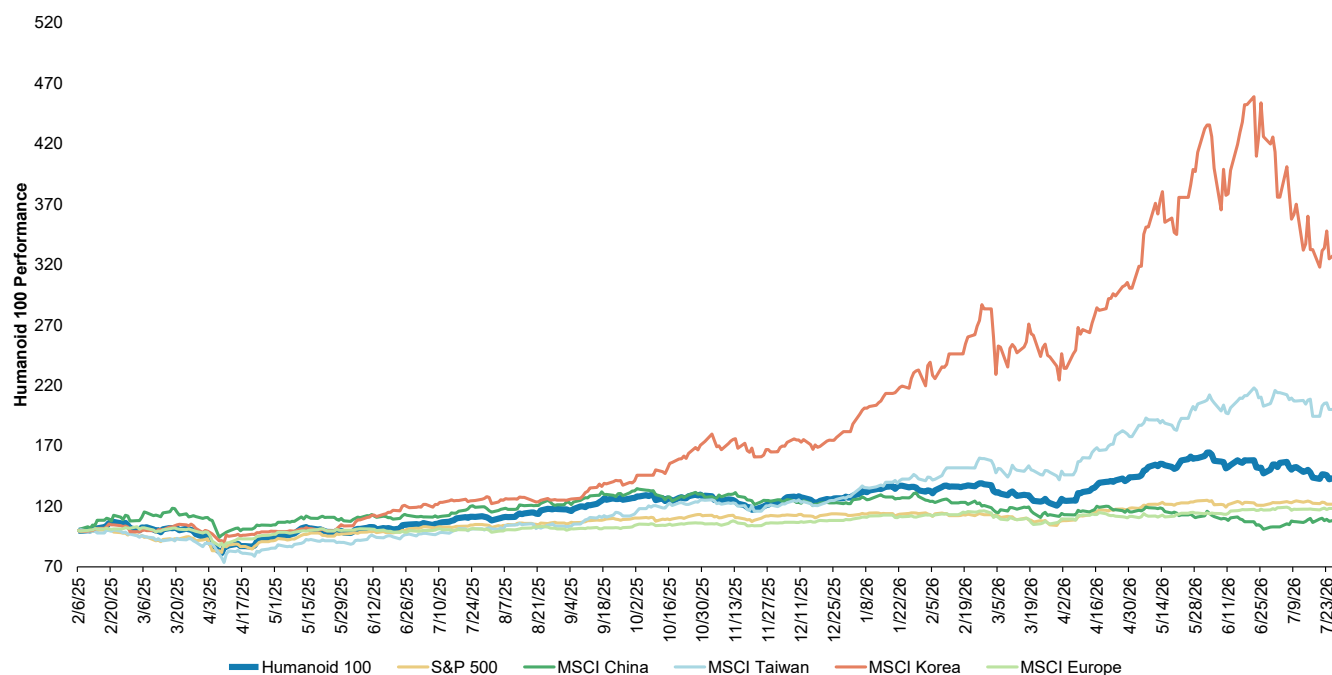


资料来源：MIR Databank、摩根士丹利研究部

如需获取我们的全球机器人模型，涵盖完整具身智能品类（不仅限于人形机器人），请联系您的摩根士丹利代表。

人形机器人100指数（Humanoid 100）更新：自2025年2月6日发布以来（在新增和剔除成分股后），等权重人形机器人100指数已上涨47%，跑赢标普500指数、明晟（MSCI）欧洲指数及明晟中国指数，但落后于明晟韩国指数及明晟台湾指数。初至今表现最佳的公司包括英特尔、LG电子、安谋控股、三星电子、帕洛阿尔托网络、意法半导体、泰瑞达、伟创力、上银科技和THK株式会社。

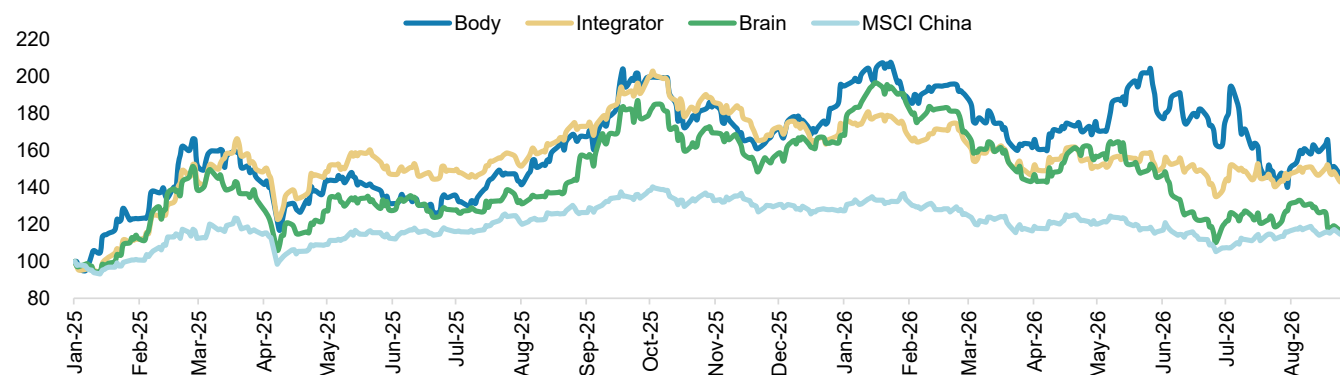
中国人形机器人价值链指数更新：截至8月28日收盘，中国人形机器人价值链指数按等权重计算月初至今下跌1.3%，大致与明晟中国指数的-0.5%一致；但年初至今累计下跌21.8%，落后于明晟中国指数（-9.1%）。

Exhibit 9: 等权重人形机器人100指数自2025年2月6日（名单设立）以来的表现

注：数据截至2026年8月28日收盘。仅计价格回报，不含股息再投资或交易成本。资料来源：FactSet、摩根士丹利研究部

Exhibit 10: 中国人形机器人价值链指数表现（等权重）

China Humanoid Value Chain - Equal-weighted Performance

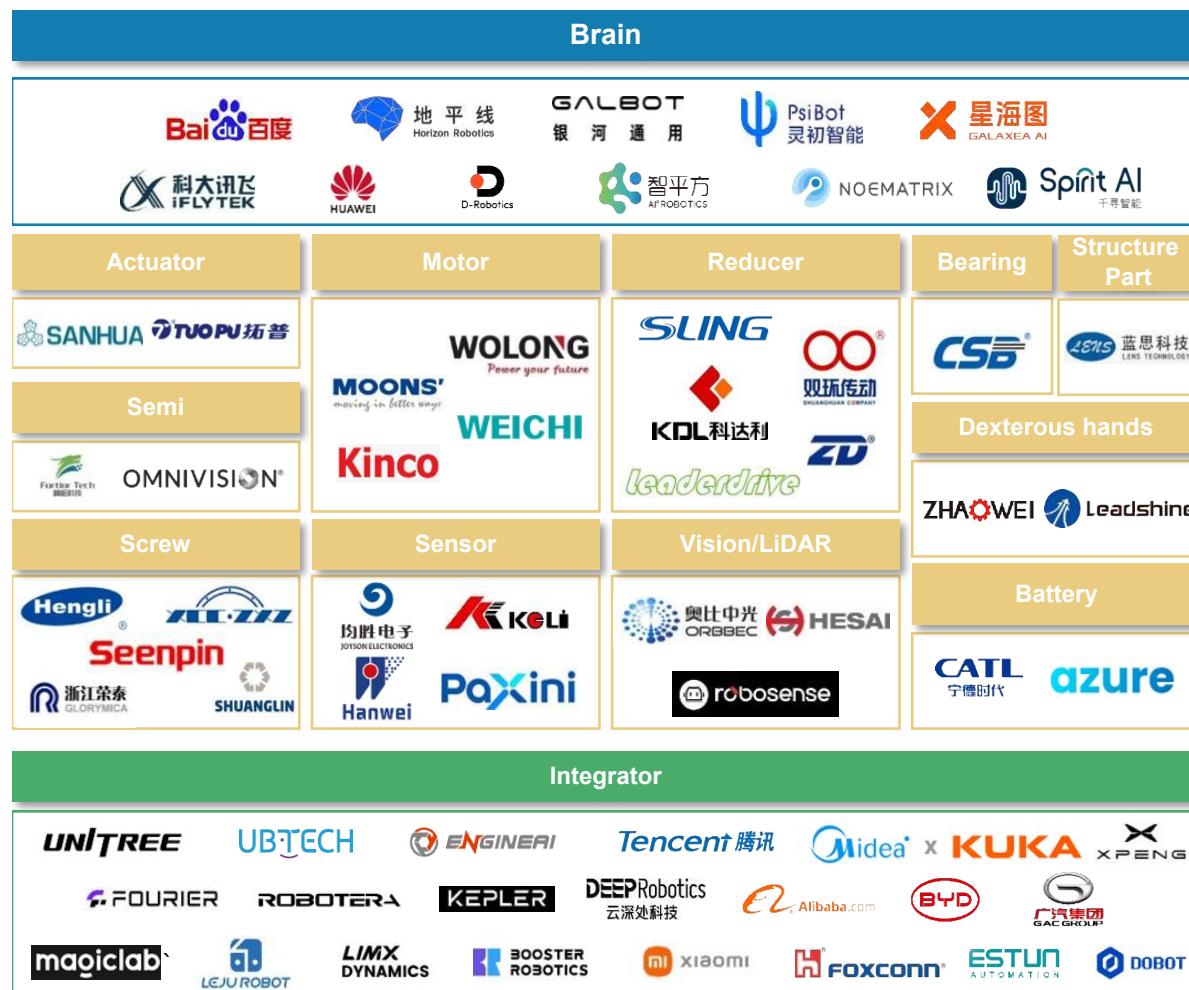


注：公司名单为下述中国人形机器人价值链名单调整前的名单。所示结果为不含股息、经纪佣金和交易成本的回报，未经审计。历史表现不保证未来结果。资料来源：FactSet（截至2026年8月28日收盘）、摩根士丹利研究部

Exhibit 11: Humanoid 100——摩根士丹利全球人形机器人赋能企业名单

Brain							Integrators	
Foundational Models	Data Science & Analytics	Cybersecurity	Simulation & Vision Software	Semis (Vision & Compute)	Semis (Designers)	Semis (Fab)	OEMs	
 百度  Alphabet  Microsoft	 Palantir  ORACLE  Microsoft	 paloalto  Microsoft	 HEXAGON  Meta  Alphabet  NVIDIA  ROBLOX  SIEMENS  unity 	 intel  NVIDIA  mobileye  Qualcomm  Ambarella 	 arm  synopsys  cadence	 TSMC  SAMSUNG  intel	 TESLA  BYD  Alibaba Group  amazon	
Body								
Actuators & Actuator Parts		Sensors	Batteries	Semis (Analog)	Body, Wiring, Thermal	Diversified Automation		
Bearings  NSK  RKC  TIMKEN  SKF  SCHAEFFLER 		Radar & Lidar  MAGNA  intel  TELEBYTE TECHNOLOGIES  APTIV  robosense	EVE Energy  SAMSUNG SAMSUNG SDI LG Energy Solution CATL	 ALLEGRO  ANALOG DEVICES  infineon RENESAS Melexis OMNIVISION TEXAS INSTRUMENTS  ST onsemi	Aluminum Castings  MAGNA  KUSHENG Wires & Connectors  Amphenol  TE APTIV Thermal  TUOPU  SANHUA	Honeywell  Rockwell Automation SIEMENS	 SAMSUNG  RAINBOW ROBOTICS  UBTECH  GAC GROUP  TOYOTA  LG  HYUNDAI  XIBOMI  ESTUN AUTOMATION  Tencent 腾讯  TERADYNE  HEXAGON  Midea  NAVER  XPENG	
Screws  NSK  Hengli  SKF  THK  HIWIN 		Magnetic  Melexis  ALLEGRO Force & Torque  Novanta  Sensata  TE Cameras & Vision Sensors  TELEBYTE TECHNOLOGIES  intel					Contract Manufacturers	
Gears / Reducers  HIWIN  TIMKEN  Nabtesco 		Encoders  Nidec  Novanta  Sensata Rare-Earths / Magnets  Lynas  MP MATERIALS  JL MAG					 FOXCONN PEGATRON  LUXSHARE ICT JABIL  flex 	

注：仅包括上市公司，非穷尽性名单。资料来源：摩根士丹利研究部

Exhibit 12: 中国人形机器人价值链（包括我们股票名单中的上市公司，见 Exhibit 25，以及非上市企业）

资料来源：摩根士丹利研究部

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(as of August 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1555	42%	450	48%	29%	767	44%
Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: China Industrials

COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/01/2026)
Chelsea Wang		
China Railway Group (601390.SS)	E (05/12/2022)	Rmb4.42
China Railway Group (0390.HK)	E (08/11/2025)	HK\$3.42
China State Construction Engineering (601668.SS)	U (08/11/2025)	Rmb4.36
Han's Laser (002008.SZ)	O (10/02/2025)	Rmb94.60
Hefei Meyer Optoelectronic Technology (002690.SZ)	E (09/08/2025)	Rmb16.29
iRay Technology Company Limited (688301.SS)	O (07/07/2026)	Rmb94.50
Neway Valve (Suzhou) Co., Ltd (603699.SS)	O (09/12/2025)	Rmb48.39
Shanghai BOCHU Electronic Technology (688188.SS)	O (08/22/2024)	Rmb98.76
Shenzhen Envicool Technology Co Ltd (002837.SZ)	O (08/19/2024)	Rmb65.90
Sheng Zhong		
Beijing Geekplus Technology Co., Ltd. (2590.HK)	O (08/07/2025)	HK\$9.06
Centre Testing International Group (300012.SZ)	E (11/18/2024)	Rmb15.16
CRRC Corp Ltd (1766.HK)	U (01/22/2026)	HK\$5.37
CRRC Corp Ltd (601766.SS)	U (01/22/2026)	Rmb6.31
DR Laser (300776.SZ)	E (12/17/2021)	Rmb109.21

Estun Automation Co Ltd (002747.SZ)	E (08/25/2026)	Rmb31.01
Haitian International Holdings Limited (1882.HK)	E (09/08/2025)	HK\$18.15
Hongfa Technology Co Ltd (600885.SS)	O (05/23/2023)	Rmb34.37
Jiangsu Guomao Reducer Co Ltd (603915.SS)	U (01/08/2025)	Rmb14.12
Jiangsu Hengli Hydraulic Co.Ltd (601100.SS)	O (05/23/2023)	Rmb101.88
Jingsheng Mechanical & Electrical Co (300316.SZ)	U (01/08/2025)	Rmb42.54
Leader Harmonious Drive Systems (688017.SS)	O (01/22/2026)	Rmb290.02
Sany Heavy Industry Co., Ltd. (600031.SS)	O (01/08/2025)	Rmb20.22
Shenzhen Inovance Technology (300124.SZ)	++	Rmb60.95
Shenzhen SC New Energy Technology Corp (300724.SZ)	U (09/08/2025)	Rmb55.40
Sinotruk (Hong Kong) Limited (3808.HK)	E (05/19/2025)	HK\$39.54
Suzhou Maxwell Technologies Co Ltd (300751.SZ)	U (09/15/2023)	Rmb207.11
Times Electric (3898.HK)	E (01/22/2026)	HK\$30.62
WeiChai Power (2338.HK)	O (03/30/2026)	HK\$31.60
WeiChai Power (000338.SZ)	O (03/30/2026)	Rmb27.35
Wuxi Autowell Technology Co Ltd (688516.SS)	U (09/08/2025)	Rmb44.32
Wuxi Lead Intelligent (300450.SZ)	O (09/08/2025)	Rmb32.81
Zhejiang Dingli Machinery Co Ltd. (603338.SS)	O (11/05/2025)	Rmb58.05
Zhejiang Hangke Technology (688006.SS)	O (09/08/2025)	Rmb21.52
Zhejiang Shuanghuan Driveline Co. Ltd. (002472.SZ)	O (08/25/2023)	Rmb35.32
Zoomlion Heavy Industry (1157.HK)	O (09/08/2025)	HK\$6.09
Zoomlion Heavy Industry (000157.SZ)	O (09/08/2025)	Rmb6.73

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* Historical prices are not split adjusted.